

# Class 04

## Filtering Data

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- **WHERE clause basics**
- **Comparison operators (=, <>, >, <, BETWEEN, LIKE)**
- **Logical operators: AND, OR, NOT**
- **IN, NOT IN, IS NULL, IS NOT NULL**

# WHERE clause

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**WHERE** clause is used to filter records in a table.  
It helps us select only those rows which meet a condition.  
Without WHERE, all rows are selected.

**Think of it like:** *"Show me only the data I need."*

## Syntax

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```
SELECT column1, column2, ...  
FROM table_name  
WHERE condition;
```



# WHERE clause

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**Students** (table)

| Id | Name   | age | city   | marks |
|----|--------|-----|--------|-------|
| 1  | Aman   | 22  | Mumbai | 60    |
| 2  | Akarsh | 23  | Bhopal | 80    |
| 3  | Aditya | 21  | Delhi  | 67    |

**Query**

```
SELECT * FROM Students  
WHERE marks > 70;
```



| Id | Name   | age | city   | marks |
|----|--------|-----|--------|-------|
| 2  | Akarsh | 23  | Bhopal | 80    |

# Comparison operators

## (=, <>, >, <, BETWEEN, LIKE)

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Comparison operators are used to **compare values** in SQL  
They help in filtering and checking conditions.



- = → Equal to
- <> or != → Not equal to
- > → Greater than
- < → Less than
- BETWEEN → Value in a range
- LIKE → Pattern matching

### Important:

Comparison operators are **not only used with WHERE** —

They can also be used in **HAVING, JOIN conditions, and CASE expressions.**

# Comparison operators

(=, <>, >, <, BETWEEN, LIKE)

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**Employees** (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

**Query**

```
SELECT * FROM Employees  
WHERE city = "Delhi";
```



| Id | Name  | age | city  | salary |
|----|-------|-----|-------|--------|
| 3  | Mohit | 21  | Delhi | 67,000 |

# Comparison operators

(=, <>, >, <, BETWEEN, LIKE)

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**Employees** (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

**Query**

```
SELECT * FROM Employees  
WHERE city <> "Delhi";
```



| Id | Name | age | city   | salary |
|----|------|-----|--------|--------|
| 1  | Aman | 22  | Bhopal | 60,000 |
| 2  | Ajay | 23  | Mumbai | 60,000 |

# Comparison operators

(=, <>, >, <, BETWEEN, LIKE)

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**Employees** (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

**Query**

```
SELECT * FROM Employees  
WHERE salary > 60,000;
```



| Id | Name  | age | city  | salary |
|----|-------|-----|-------|--------|
| 3  | Mohit | 21  | Delhi | 67,000 |

# Comparison operators

(=, <>, >, <, BETWEEN, LIKE)

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Employees (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

Query

```
SELECT * FROM Employees
WHERE age
BETWEEN 22 AND 23;
```



| Id | Name | age | city   | salary |
|----|------|-----|--------|--------|
| 1  | Aman | 22  | Bhopal | 60,000 |
| 2  | Ajay | 23  | Mumbai | 60,000 |

# Comparison operators

(=, <>, >, <, BETWEEN, LIKE)

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**Employees** (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

**Query**

```
SELECT * FROM Employees  
WHERE Name  
LIKE 'A%';
```



| Id | Name | age | city   | salary |
|----|------|-----|--------|--------|
| 1  | Aman | 22  | Bhopal | 60,000 |
| 2  | Ajay | 23  | Mumbai | 60,000 |

# IN, NOT IN, IS NULL, IS NOT NULL

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**Employees** (table)

| Id | Name  | age | city   | salary |
|----|-------|-----|--------|--------|
| 1  | Aman  | 22  | Bhopal | 60,000 |
| 2  | Ajay  | 23  | Mumbai | 60,000 |
| 3  | Mohit | 21  | Delhi  | 67,000 |

**Query**

```
SELECT * FROM Employees  
WHERE Name  
LIKE 'A%';
```



| Id | Name | age | city   | salary |
|----|------|-----|--------|--------|
| 1  | Aman | 22  | Bhopal | 60,000 |
| 2  | Ajay | 23  | Mumbai | 60,000 |